Elements of AIML
Unit 01 – Lecture 01 Notes
AI Overview, Techniques & Applications

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February 28, 2026

Contents

1 Lecture Overview **2**

2 How to use these notes (self-study) **2**

3 Learning Outcomes **2**

4 Keywords (quick definitions) **2**

5 Course roadmap (how topics connect) **3**

6 Abbreviations and terms used in this lecture **3**

7 Why AI is important now (three drivers) **3**

8 AI task types (how we frame problems) **3**

9 What is Artificial Intelligence? **4**
9.1 A practical course definition 4
9.2 Turing Test (short note) 4
9.3 Rationality (preview) 4

10 AI vs ML vs Data Science **4**
10.1 AI 5
10.2 Machine Learning (ML) 5
10.3 Data science 5

11 AI Techniques (high level) **5**
11.1 Symbolic AI (examples) 5
11.2 Statistical AI / ML (examples) 6
11.3 Hybrid systems 6
11.4 When do we choose which? 6

12 Models and criteria of success **6**
12.1 What is a model? (levels) 6
12.2 Criteria of success 6
12.3 Common failure modes 7

13 Common misconceptions (quick fixes) **7**

14 Worked example: spam filtering (rules vs learning) **7**
14.1 Rule-based approach (symbolic) 7
14.2 Learning-based approach (ML) 7
14.3 How do we measure success here? 8

15 Common pitfalls (and how to avoid them) **8**

16 Practice (with answers) **8**

17 Exit questions (with answers) **9**

18 Summary **9**

1 Lecture Overview

This lecture introduces the course and answers four practical questions:

- What do we mean by **Artificial Intelligence (AI)**?
- How is AI related to **Machine Learning (ML)** and **Data Science**?
- What are the **major technique families** used in AI systems?
- How do we define **success** for an AI system in the real world?

Sources: Rich and Knight, *Artificial Intelligence*, McGraw Hill, 2017.;Russell and Norvig, *Artificial Intelligence: A Modern Approach* (4e), Ch. 1–2 (Introduction, Intelligent Agents).;Poole and Mackworth, *Artificial Intelligence: Foundations of Computational Agents* (2e), selected sections.

2 How to use these notes (self-study)

Read the notes in order and pause at each worked example/case to ask: "What would I do next, and why?". In AI, the important skill is not memorizing definitions but connecting **problem type** → **technique** → **evaluation**. Try the practice questions before reading the answers.

3 Learning Outcomes

- Explain AI using multiple viewpoints (human-like vs rational, thinking vs acting).
- Differentiate AI vs ML vs data science using small examples.
- Identify symbolic, statistical, and hybrid AI techniques at a high level.
- Describe what a model is and list success criteria beyond accuracy.

4 Keywords (quick definitions)

- **Artificial Intelligence (AI):** systems that perceive, reason, learn, and act to achieve goals.
- **Machine Learning (ML):** methods that learn patterns from data to make predictions/decisions.
- **Model:** a simplified representation of reality used for prediction or reasoning.

- **Metric:** a quantitative measure of performance (e.g., accuracy, error, latency).
- **Generalization:** performance on unseen data drawn from the same process as training.
- **Data leakage:** unintended use of information that would not be available at prediction time.

5 Course roadmap (how topics connect)

This subject is intentionally organized from **foundations** to **methods** to **applications**:

- **Unit I (this unit):** what AI is, what problems it solves, and what success means in practice.
- **Unit II:** how to represent knowledge using logic and how to infer new facts (entailment, resolution).
- **Unit III:** how learning from data works (datasets, splits, cross-validation, dimensionality reduction).
- **Unit IV:** core ML paradigms (regression, classification, clustering, RL, semi-supervised).
- **Unit V:** where AI/ML is used and how to think responsibly about impacts and risks.

6 Abbreviations and terms used in this lecture

- **AI**: Artificial Intelligence
- **ML**: Machine Learning
- **DS**: Data Science
- **EDA**: Exploratory Data Analysis (lab focus)
- **KB**: Knowledge Base (logic-based AI; Unit II)

7 Why AI is important now (three drivers)

AI has existed for decades, but recent progress is strongly driven by:

- **Data availability:** many domains are now digital (transactions, text, images, sensors).
- **Compute availability:** GPUs/accelerators + cloud infrastructure.
- **Algorithm/tool improvements:** better optimization, architectures, libraries, and evaluation practices.

Practical takeaway: a strong AI system is rarely "just the model" — it is usually a **full pipeline** (data + model + monitoring + feedback).

8 AI task types (how we frame problems)

Many AI/ML applications reduce to a small set of task types:

- **Classification:** predict a label/class (spam vs not spam).
- **Regression:** predict a number (house price).

- **Ranking/recommendation:** order items by relevance (top videos).
- **Clustering:** group similar items (customer segments).
- **Decision making:** choose actions over time (agent behavior).

Understanding the task type helps choose the right evaluation metrics and techniques.

9 What is Artificial Intelligence?

AI can be described using different (equally valid) perspectives. A common framing (popularized in many AI textbooks) is the 2D view:

- **Thinking vs acting** (internal reasoning vs external behavior),
- **Human-like vs rational** (imitate humans vs do the "right" thing).

This gives four informal viewpoints:

- **Thinking humanly:** model human cognition (cognitive science).
- **Acting humanly:** produce human-like behavior (e.g., Turing Test style).
- **Thinking rationally:** reason correctly using logic.
- **Acting rationally:** choose actions that maximize goal achievement.

9.1 A practical course definition

For this course, a useful working definition is:

> AI is the study and engineering of systems that **perceive, reason, learn, and act** to achieve goals under uncertainty.

This definition is deliberately broad: it includes both logic-based reasoning (Unit II) and data-driven learning (Units III–IV).

9.2 Turing Test (short note)

The **Turing Test** is a historic proposal to judge whether a machine can exhibit human-like conversational behavior. It is influential, but it does not capture all goals of AI (for example, an AI system can be useful without being human-like).

9.3 Rationality (preview)

In modern AI, a central theme is **rational behavior**: choosing actions that achieve goals given limited information and resources. We study this using the **agent** framework in Unit I Lecture 02.

10 AI vs ML vs Data Science

These terms are related but not identical.

10.1 AI

AI is the *goal*: building intelligent behavior. It includes:

- **Knowledge representation and reasoning** (logic, inference, planning),
- **Learning** (ML),
- **Perception** (vision, speech),
- **Decision making** under uncertainty.

10.2 Machine Learning (ML)

ML is a *toolbox* inside AI. Instead of hand-coding rules, we learn a function from data. Example tasks:

- Regression (predict a number), classification (predict a class), clustering (group data).

ML is powerful but depends heavily on data quality and correct evaluation.

10.3 Data science

Data science focuses on extracting insights/value from data:

- Data cleaning and exploration (EDA),
- Visualization and communication,
- Experimentation and reporting.

Often, a real project includes all three: data science to understand the problem, ML to build a predictor, and AI system design to deploy decisions safely.

11 AI Techniques (high level)

At a high level, techniques can be grouped into:

- **Symbolic AI:** explicit representations (facts/rules), logic, search, planning.
- **Statistical AI / ML:** learning from data (probabilistic and optimization-based).
- **Hybrid:** combinations (very common in practice).

11.1 Symbolic AI (examples)

Symbolic AI is useful when:

- the domain has clear rules/constraints (business policies, logic constraints),
- interpretability/auditability is important.

Examples of symbolic techniques:

- **Logic and inference** (Unit II): represent facts/rules and infer consequences.
- **Search and planning**: explore actions to reach goals (e.g., route planning).

- **Constraint satisfaction**: satisfy many rules simultaneously.

11.2 Statistical AI / ML (examples)

ML is useful when:

- rules are unknown or too complex,
- labeled examples are available (supervised learning), or structure exists in data (unsupervised).

Examples:

- **Regression/classification** (Unit IV): predict values/labels.
- **Clustering** (Unit IV): discover groups.
- **Dimensionality reduction** (Unit III): compress and visualize data.

11.3 Hybrid systems

Hybrid systems combine constraints/rules with learning. Examples:

- A fraud detection model whose decisions are filtered by business rules.
- A medical model that produces suggestions but must obey safety constraints and thresholds.

11.4 When do we choose which?

A helpful rule of thumb:

- If domain rules are known and stable, symbolic methods can be clean and interpretable.
- If rules are unknown or too complex, ML can learn patterns from examples.
- If you need both (constraints + learning), hybrid approaches are effective.

12 Models and criteria of success

12.1 What is a model? (levels)

The word "model" is used in multiple ways. It helps to separate levels:

- **Conceptual model:** assumptions and simplified story of the world (whiteboard).
- **Mathematical model:** equations/objective (e.g., loss function).
- **Algorithmic model:** training and inference procedure.
- **System model:** full pipeline (data collection, preprocessing, monitoring).

12.2 Criteria of success

In real deployments, accuracy alone is insufficient. Typical success criteria include:

- **Quality:** accuracy/error rates/calibration.
- **Cost:** compute, memory, energy, annotation effort.

- **Speed:** response time (latency) and throughput.
- **Robustness:** stability to noise and distribution shifts.
- **Fairness and safety:** bias, privacy, harmful outputs.
- **Maintainability:** monitoring, retraining, drift handling.

12.3 Common failure modes

- **Bad data:** missing values, mislabeled examples, sampling bias.
- **Leakage:** using information not available at decision time.
- **Train/deploy mismatch:** users and environment change.
- **Wrong metric:** optimizing a metric that does not reflect real cost.
- **No monitoring:** silent degradation due to drift.

13 Common misconceptions (quick fixes)

- **"AI = ML"**: ML is important, but AI also includes reasoning, planning, and agents.
- **"High accuracy = success"**: metric choice must match real costs and constraints.
- **"More data always solves it"**: data quality, bias, and leakage can still break systems.
- **"Model is the product"**: production systems require pipelines, monitoring, and updates.

14 Worked example: spam filtering (rules vs learning)

Consider the task: classify emails as **spam** or **not spam**.

14.1 Rule-based approach (symbolic)

We can write human-designed rules such as:

IF subject contains "win money" AND sender is unknown THEN spam.

Strengths:

- Interpretable and easy to audit.

Weaknesses:

- Brittle: spammers adapt; rules become large and hard to maintain.

14.2 Learning-based approach (ML)

We collect labeled emails and learn a classifier. A typical feature set might include:

- word counts (bag-of-words), number of links, sender reputation, etc.

Strengths:

- Learns patterns that are hard to hand-code; can adapt with retraining.

Weaknesses:

- Requires data and careful evaluation; can inherit bias from labels/data.

14.3 How do we measure success here?

Accuracy may hide the real cost. For example:

- False positives (normal email marked as spam) can be very costly.
- False negatives (spam missed) are annoying but sometimes less costly.

So we might prioritize precision/recall or a cost-weighted metric (covered later in Unit IV).

15 Common pitfalls (and how to avoid them)

- **Treating AI, ML, and data science as the same thing:** AI is the broad goal, ML is one method family, and data science focuses on analysis/insight. Mixing them makes it harder to choose the right approach.
- **Thinking a model is "just an algorithm":** a deployed system includes data pipelines, preprocessing, monitoring, and feedback loops — the model is only one component.
- **Optimizing only accuracy:** in many real tasks (fraud, medical screening, spam) different errors have different costs; choose metrics that reflect the real objective.
- **Ignoring generalization and leakage:** evaluation must reflect unseen data; leakage can make numbers look good and then fail in practice.
- **Assuming "more complex is always better":** simpler models can be cheaper, easier to explain, and easier to maintain (important in high-stakes domains).

16 Practice (with answers)

P1. Classify the task type.

Decide whether each is classification, regression, clustering, or ranking:

- Predicting tomorrow's temperature.
- Grouping customers into segments based on purchase behavior.
- Ordering search results for a query.
- Detecting whether a transaction is fraudulent.

Answer: temperature = regression; customer segments = clustering; search results = ranking; fraud detection = classification (often imbalanced).

P2. Symbolic vs ML choice.

Give one example where symbolic AI is a good fit and one where ML is a good fit. **Answer:** symbolic: verifying compliance with explicit rules (policy constraints); ML: classifying spam emails from examples.

P3. Metric trap.

Explain why "accuracy" may be a poor metric for a medical screening model. **Answer:** because missing a true disease case (false negative) can be far more costly than a false positive; metrics like recall/sensitivity and cost-weighted measures are more appropriate.

17 Exit questions (with answers)

Q1. Give two definitions/views of AI.

Answer: One view is *acting rationally* (choose actions that best achieve goals). Another is *thinking rationally* (reason correctly using logic). Other valid views include acting/thinking humanly.

Q2. Differentiate AI vs ML vs Data Science with one example each.

Answer: AI is the broad goal (e.g., a rule-based expert system). ML is learning from data (e.g., a trained spam classifier). Data science is extracting insight (e.g., analyzing spam trends and presenting a report/dashboard).

Q3. List five criteria of success beyond accuracy.

Answer: Examples: latency, compute cost/energy, robustness to shift/noise, fairness/bias, privacy/safety, maintainability/monitoring.

Q4. Describe one failure mode caused by data issues.

Answer: Data leakage: using features that indirectly encode the label (e.g., including a post-event field that is only known after the outcome), causing unrealistically high test performance and failure in real deployment.

18 Summary

- AI is broad; ML is a key subset; data science supports understanding and communication.
- AI techniques include symbolic reasoning, ML, and hybrid approaches.
- Real success includes accuracy *and* cost, speed, robustness, fairness, safety, and maintenance.

Sources: Rich and Knight, *Artificial Intelligence*, McGraw Hill, 2017.;Russell and Norvig, *Artificial Intelligence: A Modern Approach* (4e), Ch. 1–2 (Introduction, Intelligent Agents).;Poole and Mackworth, *Artificial Intelligence: Foundations of Computational Agents* (2e), selected sections.